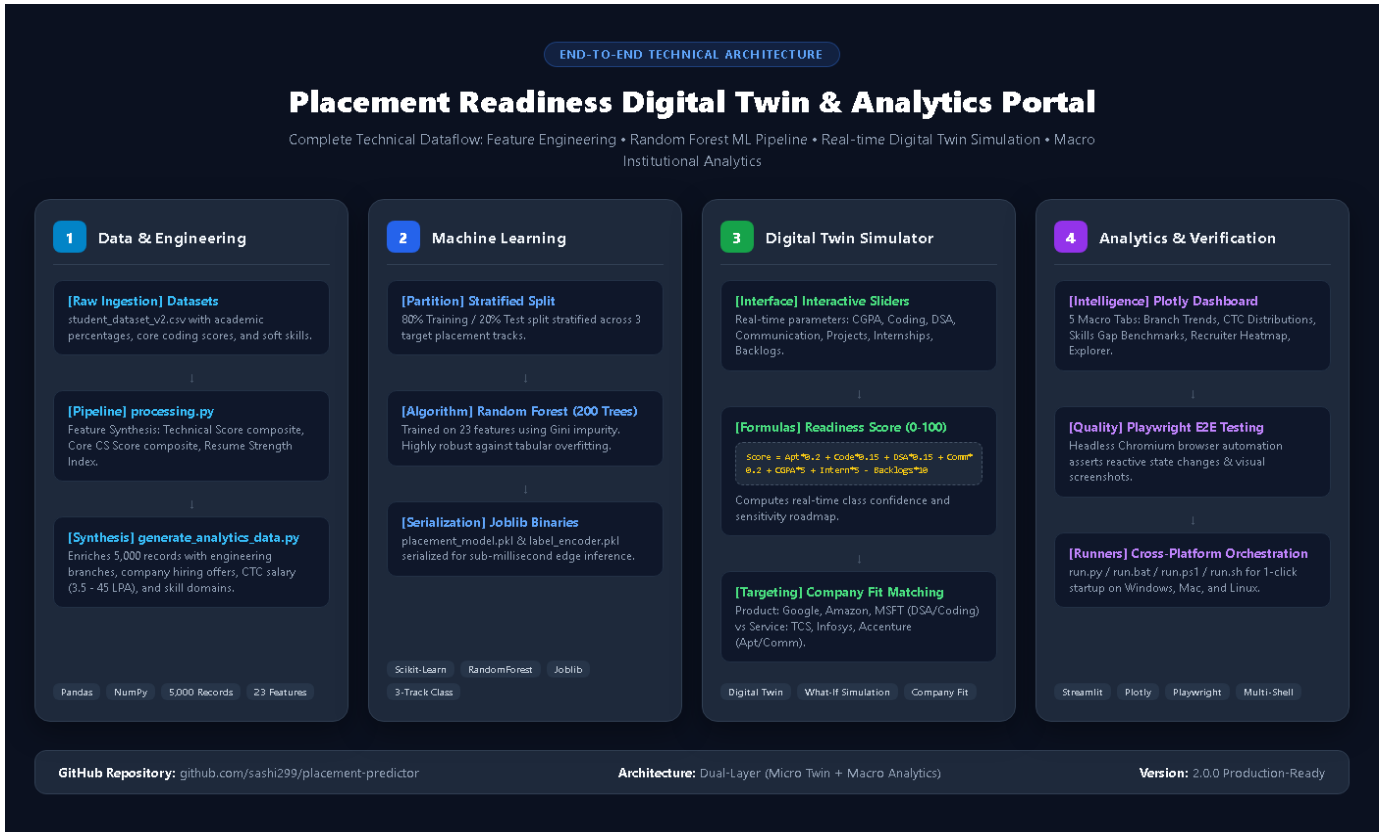


Placement Readiness Digital Twin & Analytics Portal

TECHNICAL ARCHITECTURE & DATAFLOW SPECIFICATION MANUAL



1. Data Ingestion & Feature Synthesis: Raw data (student_dataset_v2.csv) is processed via processing.py into composite metrics: Technical Score (Coding, DSA, SQL, Python avg), Core CS Score (DBMS, OS, OOPs avg), and Resume Strength. generate_analytics_data.py synthetically maps 5,000 records across 6 branches and assigns realistic CTC packages (3.5 to 45.0 LPA).

2. Machine Learning Training Pipeline: Uses a Random Forest Classifier (200 trees, Gini index) trained on 23 features with stratified 80/20 partitioning. Serialized to placement_model.pkl and label_encoder.pkl for sub-millisecond edge inference.

3. Digital Twin Simulator Engine: Evaluates student sliders in real time via Streamlit. Calculates track probability distribution and computes the normalized 0-100 Readiness Score:

$$\text{Score} = \min(100, \max(0, [\text{Apt} \times 0.2 + \text{Code} \times 0.15 + \text{DSA} \times 0.15 + \text{Comm} \times 0.2 + \text{CGPA} \times 5 + \text{Intern} \times 5 - \text{Backlogs} \times 10]))$$

Simulates targeted company fit algorithms (Google, Amazon, Microsoft, Adobe vs. TCS, Infosys, Accenture, Wipro).

4. Macro Analytics & Visualization: Powered by Plotly in analytics_dashboard.py across 5 reactive tabs: Branch Trends, Salary/CTC Statistics, Skill Benchmarks, Recruiter Heatmap, and Filtered Data Explorer with CSV download.

5. Quality Assurance & Orchestration: Validated with Playwright Chromium E2E test suites (test_analytics_playwright.py) and multi-shell launchers (run.py, run.bat, run.ps1, run.sh).

Confidential & Proprietary ? Repository: github.com/sashi299/placement-predictor